$$a_3 = \frac{a_{3-1}}{a_{3-2}+3} = \frac{a_2}{a_1+3} = \frac{1}{1+3} = \frac{1}{4}$$

Similarly the fourth term a_4 depends upon a_2 and a_3.

$$a_4 = \frac{a_{4-1}}{a_{4-2}+3} = \frac{a_3}{a_2+3} = \frac{\frac{1}{4}}{1+3} = \frac{\frac{1}{4}}{4} = \frac{1}{4}\times\frac{1}{4} = \frac{1}{16}$$

In the same way, the fifth term a_5 can be calculated as

$$a_5 = \frac{a_{5-1}}{a_{5-2}+3} = \frac{a_4}{a_3+3} = \frac{\frac{1}{16}}{\frac{1}{4}+3} = \frac{1}{16}\times\frac{4}{13} = \frac{1}{52}$$

Therefore, the first five terms of the sequence are $1, 1, \frac{1}{4}, \frac{1}{16}$ and $\frac{1}{52}$

Exercise 2.4

1. Find the next three terms of the following sequence.

   (i) $8, 24, 72, \ldots$ (ii) $5, 1, -3, \ldots$ (iii) $\frac{1}{4}, \frac{2}{9}, \frac{3}{16}, \ldots$

2. Find the first four terms of the sequences whose n^{th} terms are given by

   (i) $a_n = n^3 - 2$ (ii) $a_n = (-1)^{n+1} n(n+1)$ (iii) $a_n = 2n^2 - 6$

3. Find the nth term of the following sequences

   (i) $2, 5, 10, 17, \ldots$ (ii) $0, \frac{1}{2}, \frac{2}{3}, \ldots$ (iii) $3, 8, 13, 18, \ldots$

4. Find the indicated terms of the sequences whose n^{th} terms are given by

   (i) $a_n = \frac{5n}{n+2}$; a_6 and a_{13} (ii) $a_n = -(n^2 - 4)$; a_4 and a_{11}

5. Find a_8 and a_{15} whose n^{th} term is $a_n = \begin{cases} \frac{n^2-1}{n+3} & ; n \text{ is even}, n \in \mathbb{N} \\ \frac{n^2}{2n+1} & ; n \text{ is odd}, n \in \mathbb{N} \end{cases}$

6. If $a_1 = 1$, $a_2 = 1$ and $a_n = 2a_{n-1} + a_{n-2}$, $n \geq 3$, $n \in \mathbb{N}$, then find the first six terms of the sequence.

2.7 Arithmetic Progression

Let us begin with the following two illustrations.

Illustration 1

Make the following figures using match sticks

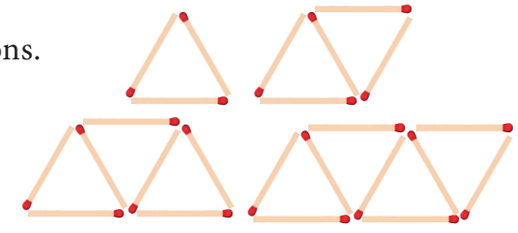


Fig.2.11

(i) How many match sticks are required for each figure? 3, 5, 7 and 9.

(ii) Can we find the difference between the successive numbers?

$5 - 3 = 7 - 5 = 9 - 7 = 2$

Therefore, the difference between successive numbers is always 2.

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